Techfest 2026-27

भारी उद्योग मंत्रालय
MINISTRY OF
HEAVY INDUSTRIES
सत्यमेव जयते

ICAT
Innovation • Service • Excellence

EV Infra & Green Mobility AI Challenge

Objective

India's automotive sector is undergoing a major transformation, driven by the adoption of electric mobility, the expansion of charging infrastructure, advances in vehicle technologies, and an increasing focus on environmentally sustainable manufacturing. National initiatives such as FAME and PM E-DRIVE have supported this transition, creating a need for robust, India-specific digital tools and datasets that can strengthen infrastructure planning and sustainability assessment.

Two critical information gaps currently constrain evidence-based decision-making. First, EV charging infrastructure information remains fragmented across multiple platforms and networks, making it difficult to assess national coverage, identify underserved regions, and align future deployment with actual vehicle populations. Second, automotive Life Cycle Assessment studies largely rely on international datasets that may not adequately reflect India's electricity mix, materials, manufacturing processes, supply chains, logistics systems, vehicle-use patterns, and recycling ecosystem.

Government Institutions, including the Ministry of Heavy Industries and its CPSEs / ABs/ Test agencies under it such as International Centre for Automotive Technology (ICAT), require scalable and transparent analytical solutions to support EV infrastructure planning, automotive sustainability assessment, policy formulation, technology evaluation, and industry engagement.

Through this innovation challenge, student teams are invited to develop practical, data-driven solutions under either of the following two challenge tracks.

Challenge Question

Participants may choose to participate in **any one** of the following two challenges.

Challenge 1: EV Infra Map India

Mapping and Integration of Nationwide EV Charger Installations Based on Vehicle Population Using VAHAN Data

India's charging infrastructure is expanding through the efforts of government agencies, public-sector entities, charge point operators, utilities, and private companies. However, information on installed chargers is distributed across multiple applications and databases. This fragmentation limits the ability to develop a unified view of charging availability and assess whether infrastructure deployment is aligned with EV adoption across regions.

Objective

Develop a data-driven national platform that:

- Integrates and maps existing public EV charging infrastructure across India.
- Uses VAHAN vehicle registration data to assess the relationship between EV population and charging availability.
- Identifies underserved areas and charging infrastructure gaps.
- Estimates future charging infrastructure requirements across different regions.
- Supports evidence-based planning by MHI, ICAT, state governments, and other relevant stakeholders.

Data Layers to Integrate

Teams may consider integrating the following broad data layers:

Data Layer	**Illustrative Sources**
EV population & vehicle registration	VAHAN database and state transport data
Existing EV charging infrastructure	Public charging applications, charger networks and open databases
Geographic & administrative boundaries	GIS platforms and open mapping sources

Road and mobility network	National highways, major roads and transport corridors
Demographic and economic activity	Population, urbanisation and commercial activity datasets
Electricity infrastructure	Publicly available grid and utility information
Supporting contextual information	Tourism, logistics, land-use and urban development data

Where live or detailed datasets are unavailable, teams may use publicly available, open-source, sample, or realistic synthetic datasets. All data sources and assumptions must be clearly documented.

Expected Deliverables

Teams selecting this challenge should submit:

- **Functional prototype or web-based platform** presenting a unified view of EV charging infrastructure.
- **Interactive national GIS map** showing existing chargers, infrastructure coverage, and identified gaps.
- **Analytical methodology document** explaining the approach used for integration, gap assessment, and estimation.
- **Charger requirement estimation and recommendation module** for selected states, districts, cities, or transport corridors.
- **Decision-support dashboard** presenting relevant indicators and planning insights.
- **Presentation or demonstration video** showcasing the solution for at least one representative geography.
- **Scalability roadmap** outlining how the solution could be expanded into a nationwide planning platform.

Challenge 2: Green Auto India

Creation of an India-Specific Life Cycle Assessment Database for the Automotive Sector

The environmental footprint of a vehicle extends beyond its operation and includes

raw material extraction, component production, vehicle manufacturing, energy consumption, transportation, maintenance, recycling, and end-of-life treatment. However, many Life Cycle Assessment studies currently depend on international or global-average datasets that may not accurately represent Indian conditions.

An indigenous automotive LCA database can support more credible environmental assessments, sustainable product development, green manufacturing, policy evaluation, ESG reporting, and technology comparison across conventional, electric, hybrid, and other emerging vehicle technologies.

Objective

Develop an India-specific automotive Life Cycle Assessment database and assessment framework that:

- Represents key materials, components, processes, energy sources, logistics activities, and end-of-life pathways relevant to India.
- Enables estimation of the environmental footprint of automotive products and technologies.
- Supports comparisons across vehicle technologies, materials, components, or manufacturing choices.
- Provides a transparent and scalable foundation for sustainability assessment.
- Supports MHI, ICAT, automotive manufacturers, suppliers, researchers, and policymakers in evidence-based decision-making.

Data Layers to Integrate

Teams may consider integrating the following broad data layers:

Data Layer	**Illustrative Sources**
Materials	Steel, aluminium, copper, plastics, composites and recycled materials
Vehicle components	Body, chassis, powertrain, battery, tyres, electronics and other major systems
Manufacturing processes	Casting, forging, machining, moulding, stamping, welding, painting and assembly
Energy sources	Electricity, renewable energy, natural gas, hydrogen and other fuels

Transportation & logistics	Road, rail, coastal shipping and other freight movements
Vehicle-use phase	Energy or fuel consumption, maintenance and operational parameters
End-of-life systems	Reuse, recycling, recovery and disposal pathways
Environmental indicators	Greenhouse gas emissions and other relevant impact categories

Teams are not expected to create an exhaustive national database during the competition. They may demonstrate the proposed framework using a defined set of materials, components, processes, or vehicle systems. All sources, boundaries, assumptions, limitations, and data-quality considerations must be clearly stated.

Expected Deliverables

Teams selecting this challenge should submit:

- **Prototype India-specific automotive LCA database** covering a clearly defined scope.
- **Database architecture and data-collection framework** demonstrating how additional datasets can be incorporated.
- **Environmental impact calculation engine** for selected automotive materials, components, or vehicle technologies.
- **Web-based dashboard or visualisation** platform presenting lifecycle results and comparisons.
- **Methodology document** explaining system boundaries, assumptions, data sources, calculations, and limitations.
- **Demonstration case study** applying the proposed database to at least one automotive product, component, material, or technology.
- **Scalability roadmap** outlining how the solution could support wider adoption by ICAT, industry, researchers, and policymakers.

Competition Format

Round 1: Concept & Roadmap Submission

The submission should be concise and include:

- **Team Information:** Team Name, Team ID, Institution, and Contact Details.
- **Methodology Document:** Explaining the team's understanding of the problem and the analytical approach they intend to use (e.g., for integration, gap assessment, and estimation under Challenge 1, or the LCA framework under Challenge 2).
- **Implementation Roadmap:** A complete, step-by-step plan explaining exactly how the team intends to build their proposed solution — including the data sources to be used, tools/technologies planned, development stages, and how they intend to arrive at the final prototype ahead of the Grand Finale.

Submission Guidelines: Online submissions for this competition must be made through the official Techfest website prior to the specified deadline. Only the Team Leader is authorized to submit on behalf of the team. Submissions made by any other team member will not be considered valid, multiple submissions from the same team will not be accepted and may lead to disqualification from the competition.
Shortlisting: There is no fixed number of teams to be shortlisted. The top-performing teams, as determined by the evaluation panel, will advance to Round 2: the Grand Finale at Techfest, IIT Bombay.

Evaluation Criteria for Round 1

Evaluation Criterion	Weight
Problem Understanding & Relevance	25%
Feasibility & Technical Soundness of Approach	25%
Innovation / Novelty of Proposed Solution	20%
Clarity & Completeness of Implementation Roadmap	20%
Planned Data Strategy (sources & integration approach)	10%

Total	100%

Round 2: Grand Finale - Live Demonstration & Presentation (Techfest, IIT Bombay)

Shortlisted teams are expected to build their proposed solution in line with the Implementation Roadmap submitted in Round 1, and present a **live demonstration of their working prototype** before an expert jury.

The presentation should include:

- A live demonstration of the working prototype/solution, covering the Expected Deliverables listed under the chosen challenge track (see Expected Deliverables).
- A brief walkthrough of how the final solution builds on or has evolved from the roadmap submitted in Round 1.

Q&A: Following the presentation, each team will go through a Q&A / jury interaction session based on a live demonstration presented.

Evaluation Criteria for Round 2

Evaluation Criterion	Weight
Technical & Analytical Robustness	25%
Data Quality & Integration	20%
Relevance & Practical Usability	20%
Innovation	15%
Scalability & Replicability	10%
Presentation & Communication	10%
Total	100%

Suggested Solution Approach (Indicative)

Teams are free to develop their own methodology. A broad solution-development approach may include:

- **Problem Definition:** Define the intended users, decision-making requirement, study boundary, and expected outputs.
- **Data Collection:** Identify, compile, and document suitable public, open-source, sample, or synthetic datasets.
- **Data Integration:** Standardise and combine datasets into a consistent analytical structure.
- **Analysis and Estimation:** Apply suitable statistical, geospatial, optimisation, LCA, or AI-based techniques.
- **Validation:** Test the model against available benchmarks, sample cases, or alternative scenarios.
- **Visualisation:** Present outputs through a clear, interactive, and non-technical user interface.
- **Recommendations:** Translate analytical results into actionable planning or sustainability insights.
- **Scalability Assessment:** Explain how the prototype can be expanded across geographies, datasets, vehicle categories, or automotive applications.

Solutions will be judged on the intelligence, transparency, and defensibility of the approach rather than access to premium or proprietary datasets.

Team Specification and Eligibility

Eligibility & Team Rules

- **Team Size: 2-5** members per team.
- **Eligibility:** Open to all undergraduate and postgraduate students. Cross-department and cross-college teams are encouraged.
- Each team may register and submit a solution under **one** challenge track only.
- An individual may not participate as a member of more than one team.

Key Dates

Event	Dates
Registration deadline	15th October 2026

Round 1 Submission deadline	30th October 2026
Round 1 Result announcement	7th November 2026
Round 2 : Finale at Techfest, IIT Bombay	16th – 18th December 2026

General Rules

1. Teams must register online at techfest.org, receiving a Team ID for future reference.
2. The organizers' and judges' decisions are final and binding. Techfest reserves the right to verify participant details.
3. All submissions must be the original work of the participating team.
4. Open-source tools, libraries, models, and public datasets may be used with appropriate attribution.
5. Any synthetic data used in the solution must be clearly identified and documented.
6. Teams must disclose the use of AI tools, third-party software, external datasets, and pre-built frameworks.
7. Finalists must be available to present and defend their solution during the Grand Finale at IIT Bombay.
8. Techfest, IIT Bombay, is not responsible for late or lost entries.
9. Solutions must be original and not protected by patents, copyrights, or publications.
10. The latest information will be available on techfest.org, with registered participants notified of changes via email.
11. All technical, procedural, and competition-related queries must be submitted directly through the Techfest Competition Portal under the Doubt section. Participants are strictly requested to regularly check the portal for answers and updates
12. Official communication for important announcements will be through the email siddhi@techfest.org. Queries submitted via email will not be entertained

Certificate Policy and Recognition

- Certificates of merit and national recognition for finalists during

the Techfest finale.
- Opportunity to showcase the solution before senior officials of the ICAT / MHI and industry leaders.
- Potential pathway for the most promising ideas to inform real municipal / ULB charging - infrastructure planning.

Prize Money

Prize money will be awarded to the **Top 3 Teams per Challenge** via NEFT, latest by May 2027.

Prize Money Submission Process

- Submission Channel: Complete the prize money form directly on the website after the final round concludes. The winners will be able to view the form through their registered email ID
- Deadline: The form must be submitted by December 31, 2026.

Details Needed for the Website Form

When filling out the form on the website, make sure you have the following bank account details ready:

1. Account Holder's Name
2. Account Number
3. Bank Name and Branch Name
4. IFSC Code